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Integrated epigenetic and genetic programming of primary human T cells

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Targeted epigenetic engineering of gene expression in cell therapies would allow programming of desirable phenotypes without many of the challenges and safety risks associated with double-strand break-based genetic editing approaches. Here, we develop an all-RNA platform for efficient, durable and multiplexed epigenetic programming in primary human T cells, stably turning endogenous genes off or on using CRISPRoff and CRISPRon epigenetic editors. We achieve epigenetic programming of diverse targeted genomic elements without the need for sustained expression of CRISPR systems. CRISPRoff-mediated gene silencing is maintained through numerous cell divisions, T cell stimulations and in vivo adoptive transfer, avoiding cytotoxicity or chromosomal abnormalities inherent to multiplexed Cas9-mediated genome editing. Lastly, we successfully combined genetic and epigenetic engineering using orthogonal CRISPR Cas12a–dCas9 systems for targeted chimeric antigen receptor (CAR) knock-in and CRISPRoff silencing of therapeutically relevant genes to improve preclinical CAR-T cell-mediated in vivo tumor control and survival.

Engineered T cells, containing transgenic T cell receptors (TCRs), chimeric antigen receptors (CARs) or other synthetic antigen receptors, are an emerging modality to treat cancer, autoimmunity and infectious diseases[1–4]. Although autologous CAR-T cells have been transformative for treating aggressive hematological malignancies, substantial advances are needed to achieve similar success in treating solid tumors and generating allogeneic cell therapies. Solid tumors demonstrate a number of challenges including immunosuppressive tumor microenvironments, physical barriers and T cell exhaustion that limit the responses of current therapies[5–7]. Allogeneic CAR-T cells must additionally overcome T cell rejection by the host immune system and toxicities such as graft-versus-host disease[8]. A number of studies have nominated genes that, in principle, can be manipulated to overcome these challenges; however, enacting these strategies in cell products remains a major challenge and will require clinically relevant, robust, nontoxic and multiplexed approaches[9–11].

CRISPR-based genome editing has become a predominant approach for engineering therapeutic T cell products. CRISPR–Cas9 can facilitate gene inactivation by introducing DNA double-strand breaks (DSBs) or stimulate precise genome editing through homology-directed repair (HDR)[12]. Base editing and prime editing can generate efficient point mutations or small insertions and deletions without introducing DSBs[13,14]. However, each of these approaches results in permanent changes to the genome and potential unintended chromosomal abnormalities induced by on-target or off-target genome editing[15–17].

An alternative strategy for modulating gene function is through programmable control of endogenous gene expression. We and

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others have shown that repurposed CRISPR proteins can perturb gene expression in T cells without the formation of DSBs or any permanent change to the genetic code[18–20]. Unfortunately, transcriptional editing approaches such as CRISPRi, CRISPRa and Cas13d require sustained expression of the CRISPR proteins to maintain control of gene expression, which largely precludes their use in therapeutic applications in T cell therapies because of immune recognition of the bacterial Cas proteins and rejection of the transplanted cells[21–23].

Recent work from our group and others has shown that heritable gene silencing can be achieved through transient expression of epigenetic effectors targeted to specific genomic loci[24–27]. In particular, we showed that CRISPRoff, an epigenetic editor protein composed of dCas9 fused to DNMT3A, DNMT3L and ZNF10 KRAB protein domains, can write an epigenetic silencing program that is persistent for over 450 cell divisions in HEK293T cells[27] when delivered as plasmid DNA. We also showed that CRISPRoff gene silencing could be reversed by CRISPRon, an epigenetic editor consisting of dCas9 fused to a TET1 catalytic domain that enables targeted erasure of DNA methylation. Importantly, epigenetic editors such as CRISPRoff and CRISPRon do not require DNA damage for their mechanisms of action, thereby eliminating the reliance on specific DNA repair outcomes and the genotoxicity, cytotoxicity and chromosomal abnormalities associated with these pathways. CRISPRoff and CRISPRon epigenome editing (epi-editing) is, thus, in principle, highly multiplexable, making complex therapeutic gene programming efforts possible for a vast array of therapeutic applications. However, this requires efficient and durable epigenetic editing in therapeutically relevant cell types such as primary human T cells.

Here, we develop an optimized, clinically compatible RNA-based epigenetic engineering platform for turning genes on and off in primary human T cells. We show that our T cell epi-editing platform is potent and durable and can be multiplexed, suggesting that this is a versatile approach for broad use in therapeutic applications. CRISPRoff gene silencing is highly specific to the intended targets, is effective in a wide range of therapeutically relevant genes and eliminates the cytotoxicity and chromosomal translocations observed with multiplexed Cas9 gene editing. Additionally, CRISPRon achieves targeted DNA demethylation of an endogenous enhancer, resulting in stable induction of *FOXP3*, a therapeutically relevant gene in human primary T cells. Lastly, we successfully couple targeted epigenome engineering with targeted CAR knock-in (KI) at the *TRAC* locus to generate epi-edited *TRAC* CAR-T cells with enhanced functionality ex vivo and in vivo in a preclinical model of adoptive cell therapy.

Results

Durable and specific silencing of endogenous genes in primary human T cells

To determine whether CRISPRoff can stably silence gene expression in human primary T cells, we designed a panel of seven mRNAs encoding the previously reported *Streptococcus pyogenes*-based CRISPRoff-V2.3 effector[27]. We compared the effects of three mRNA cap modifications (Cap1, ARCA and m^{7}g), base modifications (1-Me ps-UTP) and two codon optimization algorithms[27,28] (Supplementary Fig. 1a). The CD151 cell surface protein was selected for initial optimizations given the presence of a known CpG island (CGI) at its promoter and prior validation with CRISPRoff in HEK293T cells[27]. Each mRNA that incorporated base modifications had more *CD151* knockdown (KD) than our unoptimized mRNA, which had no 1-Me ps-UTP substitution (Supplementary Fig. 1b). We also compared codon optimization algorithms (Supplementary Fig. 1c) and mRNA cap structures (Supplementary Fig. 1d). All CRISPRoff mRNA variants demonstrated efficient CD151 KD at high mRNA concentrations, with complete silencing in 85–99% of cells and no observed cellular toxicity (Fig. 1a and Supplementary Fig. 1e). However, dose titration showed significant differences across designs in mRNA potency for gene silencing. CRISPRoff 7 mRNA, which includes 'design 1' for codon optimization, the Cap1 mRNA cap and 1-Me ps-UTP substitution, was the most potent design across mRNA concentrations, especially at low mRNA doses. We, therefore, proceeded with CRISPRoff 7 mRNA (referred to as CRISPRoff hereafter) for all subsequent experiments. We then compared CRISPRoff, CRISPRi and Cas9 mRNA activity at four different Lonza 4D nucleofector pulse codes and at four different time points (0, 2, 5 and 12 days after activation) (Supplementary Fig. 1f). Multiple pulse codes performed well for CRISPRoff KD across time points, highlighting the flexibility of this mRNA electroporation approach. We decided to use DS137 for subsequent experiments in part because it has been used previously for mRNA electroporation into T cells[29]. CRISPRoff was sufficiently efficient that we did not need to use any drug selection or cell sorting to select for CRISPRoff-positive cells.

We next examined whether CRISPRoff could initiate and maintain programmable gene silencing in primary human T cells at multiple endogenous gene targets over many cell divisions. We designed experiments to compare CRISPRoff, Cas9 and CRISPRi activity across time when delivered as mRNA to primary human T cells. We selected *CD151*, *CD55* and *CD81* for targeting as all three of these target genes contain known CGIs and are not essential for cell proliferation or survival of T cells in vitro[30]. We previously showed that CRISPRoff and

Fig. 1 | Specific and durable transcriptional silencing by CRISPRoff in primary human T cells. a, Comparison of KD efficiency of *CD151* across seven CRISPRoff mRNA designs over a series of mRNA doses. CD151 expression was assessed using flow cytometry 5 days after electroporation. We modeled CD151 positivity as a function of dose and mRNA variant and then computed a *P* value for the difference between CRISPRoff 7 and the rest of the mRNA variants using the standard error (Methods). *P* values were then adjusted using the Benjamini–Hochberg procedure. CRISPRoff 7 was the most potent CRISPRoff mRNA variant as assessed by the degree of *CD151* silencing across CRISPRoff doses ($n = 2$ donors; CRISPRoff 1, ***$P = 0.00022$; CRISPRoff 2, **$P = 0.011$; CRISPRoff 3, **$P = 0.0096$; CRISPRoff 4, **$P = 0.001$; CRISPRoff 5, **$P = 0.0044$; CRISPRoff 6, *$P = 0.04$). **b**, Comparison of Cas9 (black), CRISPRi (red) or CRISPRoff (blue) mRNA KO or KD activity on *CD151*, *CD55* and *CD81* loci over a time course of 28 days after electroporation. Black arrows along the *x* axis indicate restimulations with anti-CD2/CD3/CD28 soluble antibodies (day 9, day 18 and day 27 after electroporation) ($n = 4$ donors except on day 10, where $n = 2$ donors; mean ± s.d.). **c**, Representative flow cytometry histogram plots of *CD151* KD (or KO) by CRISPRoff, CRISPRi or Cas9 on day 5 and day 28 after electroporation. **d**,**e**, Transcriptomic assessment by RNA-seq of CRISPRoff activity and specificity upon silencing of *CD55* (**d**) or *CD81* (**e**) relative to NTC. Cells were electroporated with CRISPRoff mRNA and an sgRNA targeting *CD55* or *CD81* or NTC. Cells were harvested 28 days after electroporation for RNA extraction. Yellow dots indicate significantly downregulated DEGs and gray dots have no significance (empirical Bayes moderated statistics with Benjamini–Hochberg FDR control, adjusted $P < 0.05$; CD55 adjusted $P = 2.01 \times 10^{-5}$ and CD81 adjusted $P = 1.65 \times 10^{-4}$; $n = 2$ donors). **f**, Comparison of CpG methylation analyzed by WGBS within a 20-kb window centered on the *CD55* TSS. CGIs are depicted in green and the sgRNA targeting site is annotated. Tracks represent samples electroporated with CRISPRoff mRNA and an sgRNA targeting the *CD55* TSS or NTC for two independent donor replicates (D1 and D2). Cells were collected at 30 days after electroporation. **g**, The Manhattan plot displays DMRs between cells treated with CRISPRoff and an sgRNA targeting *CD55* or NTC and analyzed by WGBS (cells were collected at 30 days after electroporation). Red dots represent DMRs that gained DNA methylation in the targeting sgRNA samples. Blue dots represent DMRs that gained DNA methylation in NTC samples. The arrow denotes the genomic position of *CD55* ($n = 2$ donors performed in technical replicates). **h**, Day 27 transcript levels of *FAS*, *PTPN2*, *RC3H1* (Roquin 1) and *SUV39H1* relative to NTC as measured by RT–qPCR ($n = 3$ donors). **i**, Transcriptomic assessment by RNA-seq of CRISPRoff activity upon silencing of *FAS* relative to NTC. Cells were electroporated with CRISPRoff mRNA and an sgRNA targeting *FAS* or an NTC and then harvested at 7 days after electroporation for RNA extraction. The yellow dot indicates the target gene, which is significantly downregulated, and gray dots have no significance (empirical Bayes moderated statistics with Benjamini–Hochberg FDR control, adjusted $P < 0.05$; FAS adjusted $P = 3.35 \times 10^{-20}$; $n = 4$ donors).

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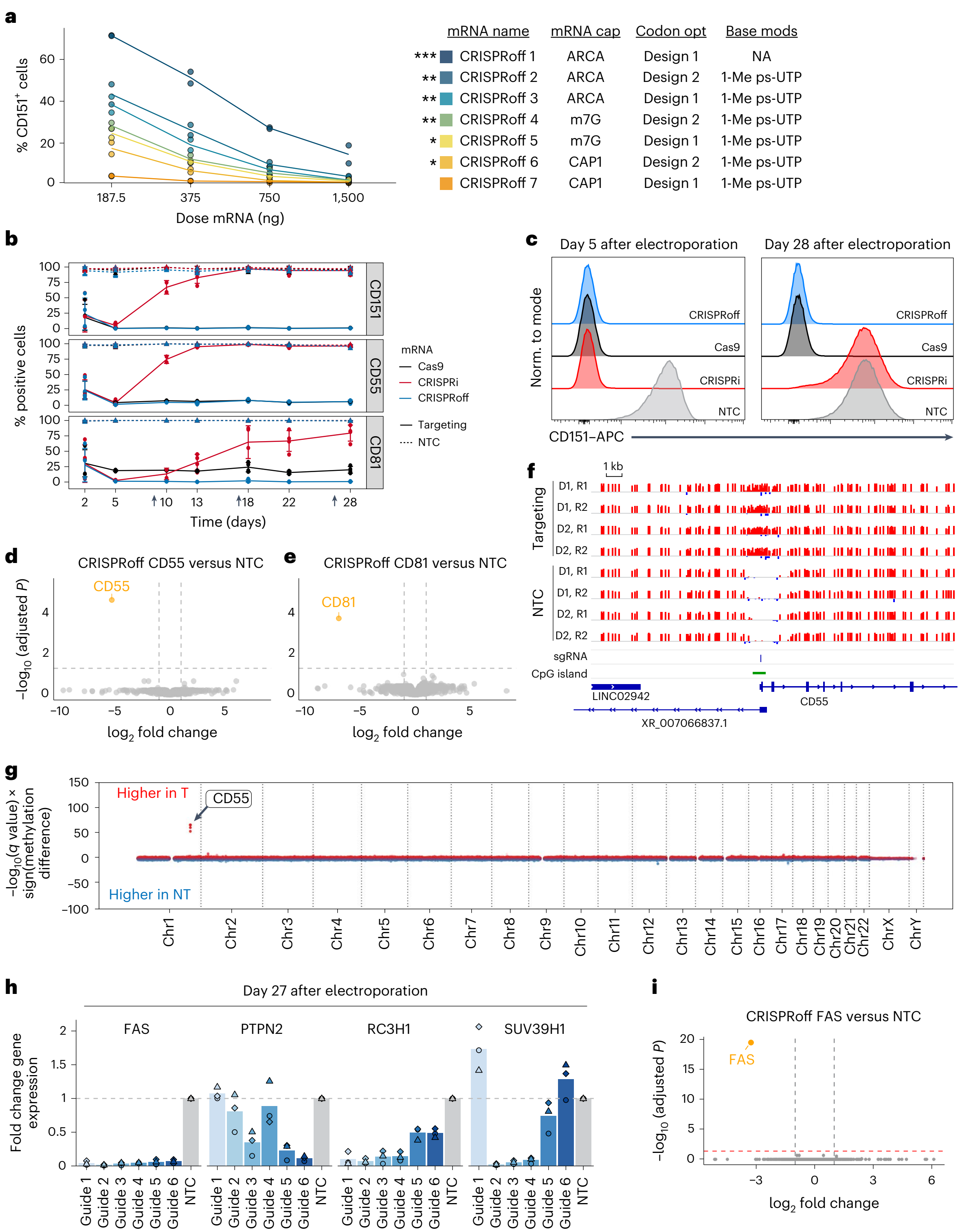
CRISPRi have equivalent design and targeting rules for optimal single guide RNA (sgRNA) activity[27]. Hereafter, throughout this study, for all CRISPRoff and CRISPRi experiments, we used the top predicted 1–6 sgRNAs[31] without prior validation in T cells. For both CRISPRoff and CRISPRi experiments, we coelectroporated a pool of three sgRNAs targeting within a 250-bp region immediately downstream of the transcription start site (TSS) of each gene or a nontargeting control sgRNA (NTC) along with CRISPRoff or CRISPRi mRNA. For Cas9 experiments, we electroporated one sgRNA predicted for optimal knockout (KO) activity or an NTC along with Cas9 mRNA[32,33]. Cell surface levels of each targeted gene's protein product were monitored by flow cytometry over a time course of 28 days. As expected, CRISPRi targeting led to transient gene silencing that was progressively lost over time, notably upon T cell restimulation using anti-CD2/CD3/CD28 soluble antibodies on day 9 (Fig. 1b). In contrast, CRISPRoff programmed durable gene silencing that was comparable to Cas9 KO for at least 28 days after electroporation with absence of cell surface expression in over 93% of cells for each gene target (Fig. 1b,c). Notably, CRISPRoff silencing persisted through three anti-CD2/CD3/CD28 soluble antibody restimulations over a 28-day time course, demonstrating that CRISPRoff gene silencing memory is stably propagated across approximately 30–80 cell divisions in vitro (Methods). RNA



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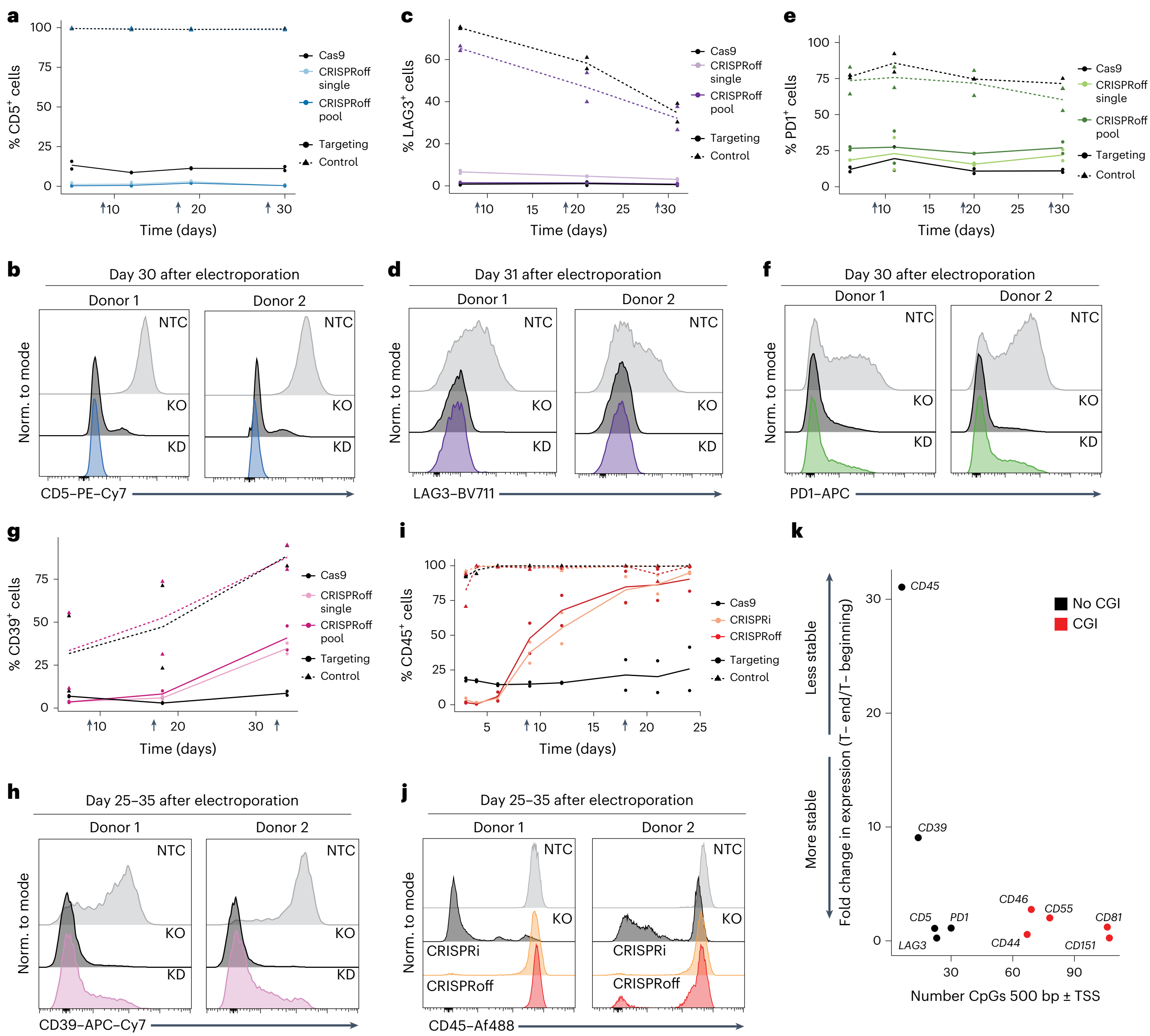

Fig. 2 | CRISPRoff silencing at genes that lack CGI annotations. **a**, Comparison of Cas9 KO (black) or CRISPRoff KD activity with either a single sgRNA (light blue) or a pool of three sgRNAs (dark blue) at *CD5* over a time course of 30 days after electroporation ($n = 2$ donors). **b**, Representative flow cytometry histogram plots of *CD5* KD (blue) or KO (black) compared to NTC (gray) at 30 days after electroporation. **c**, Comparison of Cas9 KO (black) or CRISPRoff KD activity with either a single sgRNA (light purple) or a pool of three sgRNAs (dark purple) at *LAG3* over a time course of 31 days after electroporation ($n = 2$ donors). **d**, Representative flow cytometry histogram plots of *LAG3* KD (purple) or KO (black) compared to NTC (gray) at 31 days after electroporation. **e**, Comparison of Cas9 KO (black) or CRISPRoff KD activity with either a single sgRNA (light green) or a pool of three sgRNAs (dark green) at *PDCD1* over a time course of 30 days after electroporation ($n = 2$ donors). **f**, Representative flow cytometry histogram plots of *PD1* KD (green) or KO (black) compared to NTC (gray) on day 31 after electroporation. **g**, Comparison of Cas9 KO (black) or CRISPRoff KD activity with either a single sgRNA (light pink) or a pool of three sgRNAs (dark pink) at *CD39* over a time course of 35 days after electroporation ($n = 2$ donors). **h**, Representative flow cytometry histogram plots of *CD39* KD (pink) or KO (black) compared to NTC (gray) at 31 days after electroporation. **i**, Comparison of Cas9 KO (black), CRISPRi (gold) or CRISPRoff activity with a single sgRNA (orange) targeting *CD45* over a time course of 24 days after electroporation ($n = 2$ donors). **j**, Representative flow cytometry histogram plots of KO (black), CRISPRi KD (gold) or CRISPRoff KD (orange) compared to NTC (gray) at 24 days after electroporation. In **a**,**c**,**e**,**g**,**i**, black arrows along the *x* axis indicate restimulations with anti-CD2/CD3/CD28 soluble antibodies. **k**, Fold change in surface protein expression (% positive cells) from the first time point to the last time point for cells treated with CRISPRoff mRNA and the most potent guide targeting the TSS of each respective gene. The number of CpG dinucleotides within ±500 bp of the TSS of each gene is shown on the *x* axis. Red dots indicate genes with CGI annotations in the UCSC genome browser. Black dots indicate genes that do not have a CGI annotation.

sequencing (RNA-seq) confirmed that CRISPRoff gene silencing was highly specific, with robust repression of the *CD55* or *CD81* target gene and no other differentially expressed genes (DEGs) at 28 days after electroporation (Fig. 1d,e and Supplementary Fig. 2a). Whole-genome bisulfite sequencing (WGBS) further confirmed the specificity of DNA methylation deposited at the target locus by CRISPRoff; the highest differentially methylated region (DMR) between targeting samples and NTC samples occurred at the *CD55* TSS (Fig. 1f,g).

We then tested the ability to silence therapeutically relevant genes with CGIs that are known to modulate T cell signaling or adoptive T cell function including *FAS* (ref. 34), *PTPN2* (ref. 35), *RC3H1* (Roquin 1)[36], *SUV39H1* (ref. 37), *MED12* (ref. 38) and *RASA2* (ref. 39). For *FAS*, *PTPN2*, *RC3H1* and *SUV39H1*, we electroporated CRISPRoff mRNA along with the top six predicted sgRNAs for each gene in an arrayed format alongside an NTC and then maintained cells in vitro for up to 27 days after electroporation, with restimulation using anti-CD2/CD3/CD28 soluble antibodies every 9–10 days. We also targeted each gene for KO using Cas9 as described above. Cell pellets were collected for RNA or DNA extraction and bulk RNA-seq (on day 7 after electroporation), indel analysis (on day 7 after electroporation) or qPCR (on day 27 after electroporation) was performed to measure target gene silencing or KO. We found that, for each gene, at least one and generally multiple sgRNAs could potently and durably mediate CRISPRoff silencing of target gene expression (Fig. 1h and Supplementary Fig. 2a). Cas9 gene KO was also efficient (Supplementary Fig. 2b). RNA-seq analysis enabled us to profile the biological consequences of silencing or KO of *FAS*, *MED12*, *PTPN2*, *RC3H1*, *SUV39H1* and *RASA2* and examine the specificity of CRISPRoff gene silencing (Fig. 1l and Supplementary Fig. 2c–n). *FAS* and *RC3H1* were the sole genes decreased upon CRISPRoff targeting of *FAS* and *RC3H1*, respectively. The only genes that decreased upon *SUV39H1* targeting were *SUV39H1* and *LINC02446*; *LINC02446* reduction is almost certainly a biological secondary effect of *SUV39H1* ablation because it was also decreased by *SUV39H1* KO with Cas9 and an independent sgRNA. *PTPN2* targeting also had relatively specific effects on the transcriptome with a modest number of additional downregulated genes, which could be either off-target effects or secondary effects of target KD. *RASA2* KD or *MED12* KD (targeted with a pool of three sgRNAs) had broader effects on the transcriptome. For *MED12*, to further examine CRISPRoff specificity and to determine whether observed DEGs besides the target gene were secondary transcriptional effects or potential off-targets of CRISPRoff, we compared CRISPRoff RNA-seq results to Cas9 KO RNA-seq results (Supplementary Fig. 2n). Many DEGs were shared between *MED12* KD and KO and recapitulate known biology. For example, top downregulated DEGs include *KLF2, CCR7* and *IL7R*, which are all expected biological secondary effects of *MED12* loss on the basis of past observations from our group and others[38,40] (Supplementary Fig. 2l–n). To further investigate CRISPRoff specificity, we analyzed gene expression changes for neighboring genes within a 100-kb window around each target gene (to look for on-target, off-gene effects). We also examined gene expression changes for predicted off-target sgRNA-binding sites (https://www.idtdna.com/site/order/designtool/index/CRISPR_SEQUENCE) within ±1 kb of a gene's TSS, according to previously established rules for CRISPRoff activity and specificity[27]. Only one gene across 151 putative off-target sites for all genes targeted showed evidence of potential CRISPRoff off-target activity (Supplementary Fig. 3a–f). Specifically, in *RASA2*-KD samples, we observed decreased expression of *LARP1B*. However, further work would be necessary to determine whether LARP1B is a true off-target or an indirect effect associated with *RASA2* KD. In summary, CRISPRoff gene silencing proved programmable, efficient, specific and durable at many endogenous genes in primary human T cells.

CRISPRoff silencing at genes that lack CGI annotations

In addition to its activity at CGIs, CRISPRoff was previously shown in HEK293T cells to allow durable silencing (in a DNA methylation-dependent manner) of genes lacking a CGI[27]. To explore whether this biology extends to primary T cells, we attempted to use CRISPRoff for durable silencing of five genes lacking CGIs: *CD5*, *LAG3*, *PDCD1*, *ENTPD1* (CD39) and *PTPRC* (CD45). Each of these genes encodes a cell surface protein that is important for sensing external stimuli[41], signaling[42–44] and function[45,46] in T cells. We first electroporated CRISPRoff mRNA and an individual sgRNA or a pool of three sgRNAs targeting the TSS of *CD5*, *LAG3*, *PDCD1* or *CD39* or a single NTC sgRNA. As a control, we also electroporated Cas9 mRNA and one KO sgRNA targeting each of these genes or an NTC sgRNA. We then measured cell surface levels of each target protein at time points up to 35 days after electroporation. For each time point, cells were restimulated using anti-CD2/CD3/CD28 soluble antibodies and cell surface expression was measured 24 h later using flow cytometry. For the non-CGI genes targeted, we observed stable, partially stable or unstable silencing over the course of ~30 days. For *CD5* and *LAG3*, CRISPRoff silencing was comparable to or even more efficient than Cas9 KO. At 30 days after electroporation, the CRISPRoff pooled sgRNA conditions for *CD5* and *LAG3* remained up to 99.5% and 99.1% silenced, respectively (Fig. 2a–d). While we observed some differences between the efficiency of PD1 silencing using CRISPRoff between $CD4^+$ and $CD8^+$ T cells, PD1 remained stably silenced out to 30 days after electroporation in most cells across of a bulk population (78.15% *PD1*-KD cells versus ~90% *PD1*-KO cells) (Fig. 2e,f and Supplementary Fig. 4). CD39 exhibited partially stable silencing, with a fraction of cells regaining CD39 expression with time, although most cells (53%) remained $CD39^-$ negative compared to the NTC on day 35 (Fig. 2g,h). We then evaluated CRISPRoff or CRISPRi mRNA with a single sgRNA targeting the *CD45* TSS or an NTC sgRNA in comparison to Cas9 mRNA with a single sgRNA targeting protein coding exon 2 or an NTC sgRNA. Initially, CRISPRoff, CRISPRi and Cas9 all showed efficient ablation of CD45 (~99% for KD and ~85% for KO); however, by 7 days after electroporation, both CRISPRoff and CRISPRi effects diminished until reaching the levels of the NTC by 24 days after electroporation (Fig. 2i,j). Taken together, we observed a range of how effectively and durably non-CGI genes can be silenced and found that some genes with particularly low levels of CpG dinucleotides around the TSS remain challenging to stably silence (Fig. 2k). Further work is needed to elucidate rules for governing stable silencing at non-CGI genes in primary human T cells. Nonetheless, we demonstrate that expression from CGI and non-CGI genes can be stably silenced in primary human T cells through transient delivery of CRISPRoff.

Durable multiplexed gene silencing

Epi-editing can modulate gene expression without inducing DSBs, in contrast to Cas9 nuclease targeting. This feature could offer important advantages in the context of multiplexed gene targeting approaches, as genome editing with nuclease-active Cas9 can result in translocations or chromosomal loss, which both have potential to be detrimental to cell proliferation, cell survival and perhaps the safety of therapeutic cell products[9,47–49]. To explore this approach, we simultaneously targeted sets of three, four or five nonessential genes with CRISPRoff or nuclease-active Cas9. Targeting multiple genes with nuclease-active Cas9 resulted in substantial cellular toxicity in human T cells, which may be attributed to the multiple DSBs generated by this approach (Fig. 3a). In contrast, targeting three, four or five genes for silencing with CRISPRoff resulted in minimal to no observable cellular toxicity compared to electroporation alone at either a high or low dose of mRNA (Fig. 3a and Supplementary Fig. 5a,b). CRISPRoff multiplexed epi-editing averaged across four donors was efficient and durable out to 30 days after electroporation with combined silencing of three, four and five target genes at 93.5%, 82.4% and 65.8%, respectively (Fig. 3b,c,e). For some of these multiplexed gene combinations, silencing was marginally improved by increasing the dose CRISPRoff mRNA (Supplementary Fig. 5c–f). For these multiplexing experiments, we confirmed efficient CRISPRoff gene silencing for each gene individually with greater than 95% KD when targeting one sgRNA to the TSS (Supplementary Fig. 5g). We also achieved efficient, multiplexed gene silencing of three potentially therapeutically relevant genes (*FAS*, *RC3H1* and *SUV39H1*), knocking down each target transcript by ~80% (Fig. 3d).We anticipate that durable multiplexing silencing could be further improved by empirically testing individual sgRNAs targeting genes of interest at lower doses to identify sgRNAs with optimized potency for use in multiplexed combinations.

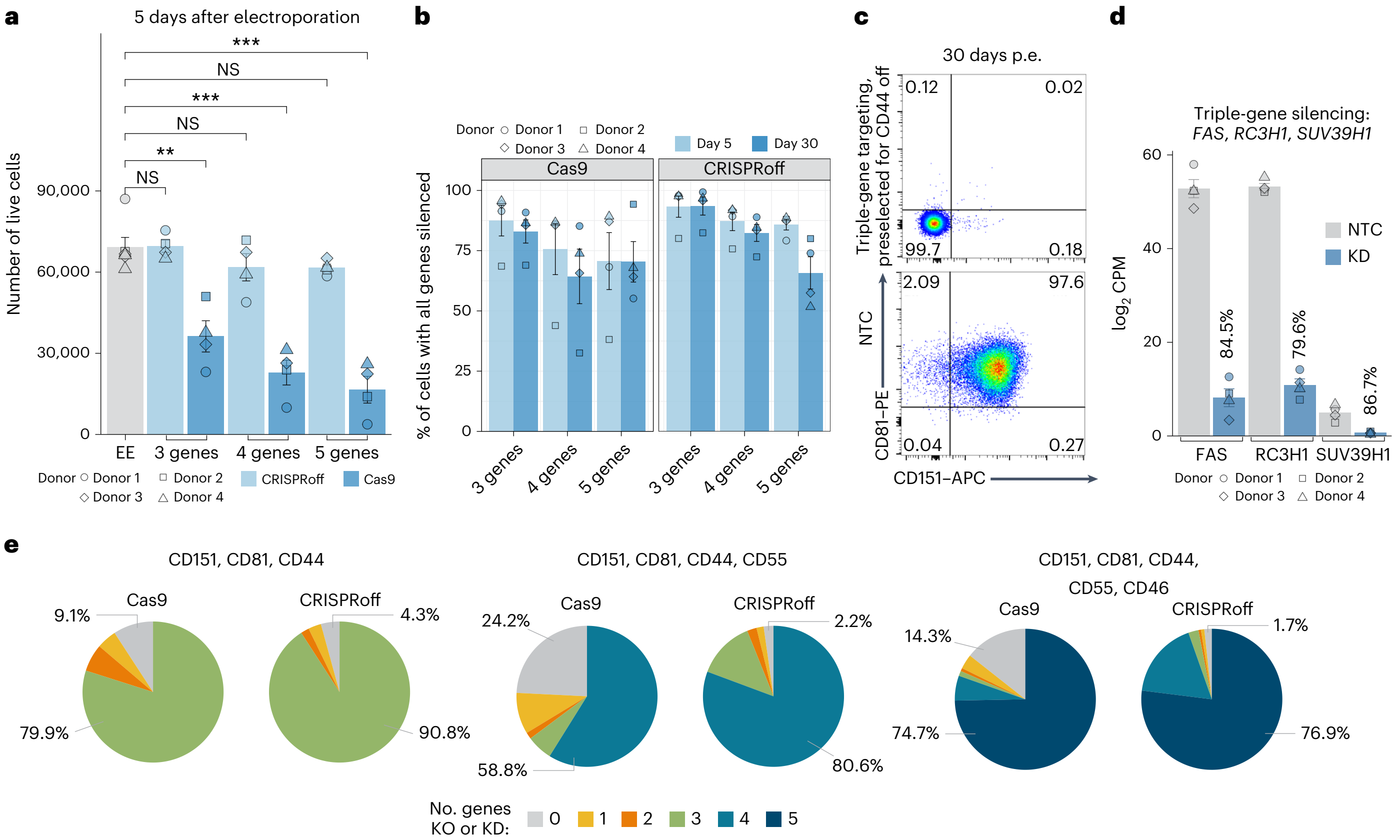

Fig. 3 | Durable multiplexed gene silencing. **a**, Graph showing the number of live T cells following cell editing by CRISPRoff and Cas9 when targeting either three, four or five genes simultaneously as compared to an empty electroporation (EE) control. Live-cell counts were measured 5 days after electroporation ($n = 4$ donors; mean ± s.e.m.; two-sided Welch's t-test: Cas9, three genes, $^{**}P = 0.004$; Cas9, four genes, $^{***}P = 0.00015$; Cas9, five genes, $^{***}P = 0.00013$; NS, not significant). **b**, Plot comparing CRISPRoff versus Cas9 multiplexed gene silencing efficiency targeting three genes (*CD151*, *CD81* and *CD44*), four genes (*CD151*, *CD81*, *CD44* and *CD55*) or five genes (*CD151*, *CD81*, *CD44*, *CD55* and *CD46*) at 5 days and 30 days after electroporation. The percentage of cells with all genes silenced was calculated from flow cytometry analysis ($n = 4$ donors; mean ± s.e.m.). **c**, A representative flow plot of cells targeted for triple-gene (*CD151*, *CD81* and *CD44*) silencing (top) or NTC (bottom). Cells were analyzed at 30 days after electroporation. Top, cells were first gated on CD44-silenced cells and the represented population shows CD81 and CD151 silencing. **d**, An RNA-seq log2 CPM (normalized counts per million) plot showing triple-target-gene KD in cells electroporated with CRISPRoff mRNA and sgRNAs targeting *FAS*, *RC3H1* and *SUV39H1* or an NTC sgRNA. Cells were collected for RNA-seq 7 days after electroporation ($n = 4$ donors; mean ± s.e.m.). **e**, Pie charts depicting the outcomes of three-gene, four-gene or five-gene silencing or KO shown in **b**. Data are representative of one donor.

CRISPRon can target an enhancer region in primary human T cells

Epi-editing would be further enabled by technology to stably activate target chromatin sites, in addition to the silencing capability demonstrated above. We previously developed CRISPRon, a deactivated Cas9 enzyme fused to the TET1 DNA demethylase catalytic domain, to remove DNA methylation from a targeted locus and induce gene expression[27]. However, this prior work with CRISPRon was performed in immortalized cell lines and only demonstrated reactivation of genes through demethylation of TSSs that were previously silenced by CRISPRoff. Here, we set out to extend the capabilities in three important ways: (1) enable use in primary cells; (2) demonstrate activation of a genomic element that is naturally DNA methylated and epigenetically silenced; and (3) extend the use of epi-editing from TSSs to an enhancer element. We generated three variants of CRISPRon mRNA (TETv3, TETv4 and TETv5), which differed in the linker length used between TET1 and dCas9, with the cap structure and base modifications we optimized for CRISPRoff (Fig. 4a, Supplementary Fig. 6a). To test whether CRISPRon could modulate clinically relevant gene expression in primary human T cells, we turned toward the *FOXP3* (forkhead box P3) locus. FOXP3 is a transcription factor necessary for establishing immune-suppressive regulatory T cells (Tregs). Tregs are essential for immune homeostasis through establishing tolerance against self-antigens, limiting inflammation and aiding in tissue repair. Tregs are usually defined by high and sustained FOXP3 expression, whereas conventional $CD4^+$ T cells (Tconvs) only transiently express FOXP3 upon activation. The difference in FOXP3 expression between cell types can be partly attributed to an intronic enhancer that harbors the Treg-specific demethylated region (TSDR), which is completely demethylated in Tregs but remains highly methylated in Tconvs[50] (Fig. 4b). We hypothesized that targeting the TSDR in Tconvs with CRISPRon could remove endogenous repressive DNA methylation at this regulatory element, resulting in constitutive FOXP3 expression[51]. We isolated CD4 + $CD25^{low}$ Tconvs from two to four human donors and stimulated cells with anti-CD2/CD3/CD28 soluble antibodies. At 2 days after stimulation, we electroporated CRISPRon-TETv3 and five individual sgRNAs targeting the TSDR, *FOXP3* TSS or *AAVS1* safe harbor locus as a control (Supplementary Fig. 6b). We then measured FOXP3 expression at 9 days after initial stimulation, at which point Tconvs have entered a resting state and, thus, should express low amounts of *FOXP3*. Only sgRNAs targeting the TSDR increased *FOXP3* expression, while targeting the TSS did not increase *FOXP3* relative to control sgRNAs (Supplementary Fig. 6b). At 48 h after restimulation, when control Tconvs transiently express *FOXP3*, cells treated with CRISPRon-TETv3 targeting the TSDR expressed higher levels of *FOXP3* than AAVS1-targeted control cells

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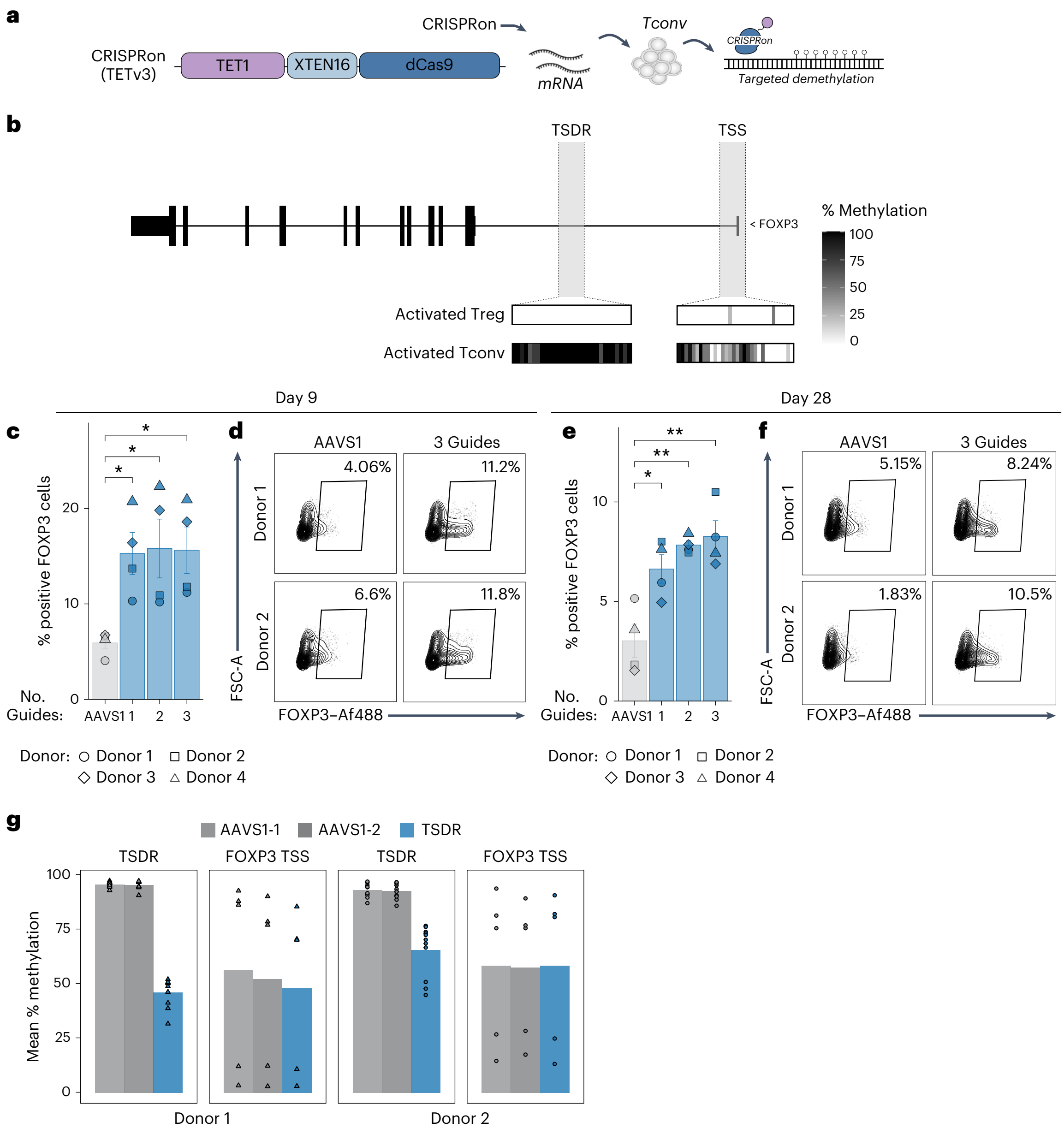

Fig. 4 | CRISPRon can target enhancer regions in primary human T cells. **a**, Schematic of CRISPRon-TETv3 mRNA, which consists of TET1 catalytic domain fused to dCas9, and work flow of targeted demethylation using mRNA electroporation. **b**, The *FOXP3* gene body with the TSDR and TSS highlighted in gray. The heat map indicates the percent methylation of individual CpGs as measured by PBAT-seq[50] across the TSDR or TSS between activated Tregs and Tconvs. Each bar in the heat map represents an individual CpG. Data are representative of one donor. **c**, Percentage of FOXP3$^+$ Tconv cells after epi-editing with CRISPRon-TETv3 mRNA and 1–3 sgRNAs targeting the TSDR (blue) or AAVS1 control region (gray) as measured by flow cytometry at 9 days after electroporation (n = 4 donors per condition; mean ± s.e.m.; two-sided Welch's t-test: for one guide targeting the TSDR, *P = 0.02; for two guides, *P = 0.046; for three guides, *P = 0.024). **d**, Representative flow plots from day 9 after electroporation depicting *FOXP3* expression after epi-editing with CRISPRon-TETv3 targeting the TSDR with a pool of three sgRNAs or an AAVS1 control. **e**, Percentage of FOXP3$^+$ Tconv cells after epi-editing with CRISPRon-TETv3 mRNA targeting the TSDR with 1–3 sgRNAs or an AAVS1 control as measured by flow cytometry at 28 days after initial activation (n = 4 donors per condition; mean ± s.e.m.; two-sided Welch's t-test: for one guide targeting the TSDR, *P = 0.018; for two guides, **P = 0.0083; for three guides, **P = 0.004). **f**, Representative flow cytometry histograms of FOXP3 median fluorescence intensity for CRISPRon-TETv3 and a pool of three sgRNAs targeting the TSDR or an AAVS1 control at 28 days after electroporation. **g**, Mean percentage of methylation across all CpGs assayed per targeted region (TSDR or TSS). Each triangle (donor 1) or circle (donor 2) represents an individual CpG.

(Supplementary Fig. 6c). We then took the top three performing guides targeting the TSDR from this initial experiment (guide 1, guide 3 and guide 4) and tested them either as individual sgRNAs or as pools of two or three sgRNAs with each CRISPRon mRNA variant. We observed that targeting CRISPRon to the FOXP3 TSDR resulted in an increased fraction of FOXP3$^+$ cells across multiple donors, CRISPRon designs and sgRNA number relative to the CRISPRon AAVS1 controls (Fig. 4c,d and Supplementary Fig. 6d).

We maintained the TSDR targeting conditions and AAVS1 control cells in culture to assess the stability and persistence of FOXP3 expression over time, restimulating cells with anti-CD2/CD3/CD28 soluble antibodies every 9–11 days. On day 28 after initial stimulation, Tconvs were collected for flow cytometry and we observed that *FOXP3* expression was stably upregulated in a population of cells over weeks in vitro (Fig. 4e,f and Supplementary Fig. 6e). Targeted bisulfite sequencing confirmed reduced methylation at the TSDR in TSDR-targeted cells even though the bulk bisulfite sequencing was performed on a heterogeneous population of cells, as we did not sort for FOXP3$^+$ expression before bisulfite sequencing (Fig. 4g). As expected, methylation at the *FOXP3* TSS did not change when targeting the TSDR (Fig. 4g) nor did it change at other Treg associated genes (*IL2RA* and *IKZF2*) (Supplementary Fig. 6f,g). Our optimized CRISPRon results contrast with a previous effort to demethylate the TSDR locus in a targeted manner in Tconvs, which showed rapid remethylation of the locus at late time points in culture, even when FOXP3$^-$ negative clones were isolated[51]. Here, without sorting or clonal isolation, we were able to achieve a significantly increased fraction of FOXP3$^+$ cells across donors after 28 days in culture with CRISPRon targeting the TSDR, as compared to the *AAVS1* sgRNA control. These results establish CRISPRon as a powerful tool to control expression of an important endogenous gene expression through enhancer targeting with potential to be applied toward next-generation cell therapies.

CAR-T cell enhancement with genetic and epigenetic engineering

Having established a robust toolbox for epigenome engineering in primary T cells, we applied it to enhance immune cell therapy function in a preclinical model of cancer. We aimed to use CRISPRoff to enhance CAR-T cell function by simultaneous targeted genomic integration of a CAR (or other antigen receptor) transgene along with targeted epigenetic silencing using CRISPRoff. Targeted insertion of a CAR to the endogenous TCRα constant (*TRAC*) locus using CRISPR–Cas9 genome editing can enhance T cell potency by placing CAR expression under the regulated and dynamic control of the endogenous TCRα promoter, limiting exhaustion and dysfunction[52]. This approach offers potential functional, safety and cost benefits over current lentiviral and gammaretroviral transduction methods. Our rationale for combining this approach with CRISPRoff is based on the clinical observation that introduction of a CAR alone is insufficient to achieve durable responses or cures for most cancers. Our group and many others have identified additional genes that can be disrupted to further enhance CAR-T cell function in challenging tumor microenvironments, which we reasoned would be appropriate targets for epi-editing[38,53–55]. In particular, we discovered that *RASA2* ablation promotes T cell function across a variety of immunosuppressive conditions, improving antigen sensitivity and durable effector function[39].

We reasoned that combining *TRAC* CAR KI with CRISPRoff-mediated silencing of additional targets could boost CAR-T cell function while avoiding translocations and other genotoxic events seen with prior multiplexed KO approaches[56]. We first explored an orthogonal Cas approach using *Acidaminococcus* sp. Cas12a (AsCas12a) ribonucleoproteins (RNPs) for targeted CAR KI in combination with stable epigenetic silencing of RASA2 using the *S. pyogenes* dCas9-based CRISPRoff system[57,58]. AsCas12a was precomplexed with a *TRAC* CRISPR RNA (crRNA) and coelectroporated with CRISPRoff mRNA and 1–3 sgRNAs targeting *RASA2*. Following electroporation, cells were transduced with an adeno-associated virus (AAV) HDR template (HDRT) containing a CD19-specific 28z CAR transgene flanked by *TRAC* locus homology arms, which serves as the HDR donor for KI (Fig. 5a). The addition of CRISPRoff mRNA and sgRNAs targeting *RASA2* did not reduce CAR KI efficiency or yield (Fig. 5b). Likewise, CRISPRoff exhibited robust *RASA2* silencing activity, similar in cells with or without integration of a CAR (Fig. 5c,d). In addition, we tested a fully nonviral approach for CAR KI using Cas9-target-site-modified single-stranded DNA (ssDNA) templates that were previously adapted for good manufacturing practice (GMP)[59]. We observed that using the same species of Cas9 and dCas9 for KI and KD resulted in translocations between *TRAC* and *RASA2* and less efficient CAR KI, presumably because of sgRNA swapping that led to Cas9-mediated DSBs at both loci (Supplementary Fig. 7a,b)[60]. To address guide swapping, we tested truncated sgRNAs (16-bp protospacer) for CRISPRoff targeting *RASA2* with the goal of retaining dCas9 binding and transcriptional control while eliminating Cas9 nuclease activity[61]. Truncated sgRNAs ameliorated *RASA2*:*TRAC* translocations (Supplementary Fig. 7a), retained efficient CAR KI (Supplementary Fig. 7b) and maintained silencing activity, albeit to a lesser extent than did full-length sgRNAs (Supplementary Fig. 7c–e). Taken together, we developed multiple approaches that could be made GMP-compatible that combine targeted CRISPR KI and programmable epigenome engineering.

We then tested the durability and functional effect of *RASA2* silencing by CRISPRoff in CAR-T cells through an in vitro repetitive stimulation assay with *RASA2* KD using a pool of three full-length sgRNAs or NTC. Experiments were performed with AsCas12a-based KI using an AAV HDRT given the more stable epigenome engineering observed with the full-length CRISPRoff sgRNAs. Most control CAR$^+$ cells displayed an immunophenotype consistent with a T memory stem cell population at 7 days after electroporation on the basis of CD45RA and CD62L expression, although *RASA2*-KD cells shifted slightly to a more T effector-like population, consistent with previous reports[39] (Fig. 5e,f). *RASA2*-silenced CAR-T cells were cocultured with CD19-expressing tumor cells at multiple effector-to-target (E:T) ratios repeatedly every 48 h (Fig. 5g and Methods). Consistent with previous reports, this repetitive stimulation assay caused control-edited CAR-T cells (treated with CRISPRoff and an NTC sgRNA) to decline progressively in their ability to control cancer cells by the last stimulation (Fig. 5h). *RASA2*-silenced CAR-T cells continued to kill target cells efficiently after five rounds of stimulation (Fig. 5h), consistent with the reported behavior of *RASA2*-KO CAR-T cells[39]. *RASA2* remained stably silenced in RASA2-targeted CAR-T cells isolated after the last stimulation, confirmed by western blot (Fig. 5i). Nonviral Cas9-based KI cells that had *RASA2* silenced with a pool of three truncated sgRNAs also performed better than control-edited cells in a repetitive stimulation assay (Supplementary Fig. 7f).

We next examined the stability of CRISPRoff-induced silencing in CAR-T cells when transferred in vivo. As *RASA2* silencing confers CAR-T cells with an in vivo fitness advantage over control-edited CAR-T cells, we instead chose to target *CD151*, which has no known role in T cell fitness in vivo. First, A375 melanoma cells engineered to express CD19 were engrafted in the flanks of NSG mice. Epi-edited T cells were engineered as previously described with an AsCas12a RNP precomplexed with a *TRAC* crRNA for *CD19*-CAR KI and CRISPRoff mRNA coelectroporated with a pool of three sgRNAs targeting the *CD151* TSS or an NTC (Supplementary Fig. 8a). Epi-edited or control-edited CAR-T cells were cultured in vitro for 1 week after electroporation and then transferred in vivo through tail-vein injection 1 week after A375 engraftment. At 14 days after CAR-T cell transfer, tumors and spleens were isolated from mice and CAR-T cell *CD151* expression was assessed by flow cytometry. Compared to NTC CAR-T cells, *CD151* targeted CAR-T cells obtained from the tumor and spleen retained highly efficient *CD151* KD, suggesting that CRISPRoff silencing is stable upon transfer in vivo and tumor-antigen recognition (Supplementary Fig. 8b–d).

Lastly, we tested the functional efficacy of *RASA2*-silenced CAR-T cells in vivo. As described above, we generated *RASA2*-silenced *TRAC* CAR-T using AsCas12a KI with an AAV template and CRISPRoff with a pool of three full-length sgRNAs targeting RASA2 or a single NTC sgRNA (Supplementary Fig. 9a,b). NSG mice were injected intravenously with Nalm6 leukemia cells and, 4 days later, injected with *RASA2*-silenced *TRAC* CD19-28z CAR-T cells, control *TRAC* CD19-28z CAR-T cells (treated with CRISPRoff and a single NTC sgRNA) or *TRAC*-KO T cells through the tail vein (Supplementary Fig. 9c). Tumor burden was monitored by bioluminescence imaging (BLI) for ~40 days. We found that *RASA2*-silenced CAR-T cells had a significant advantage over NTC CAR-T cells at controlling tumor burden in vivo in cohorts of mice treated with cells from multiple human donors (Fig. 5j and Supplementary Fig. 9d). Mice treated with *RASA2*-silenced *TRAC* CAR-T cells also had significantly extended survival relative NTC *TRAC* CAR-T cells (Fig. 5k,l). Taken together, these data suggest that

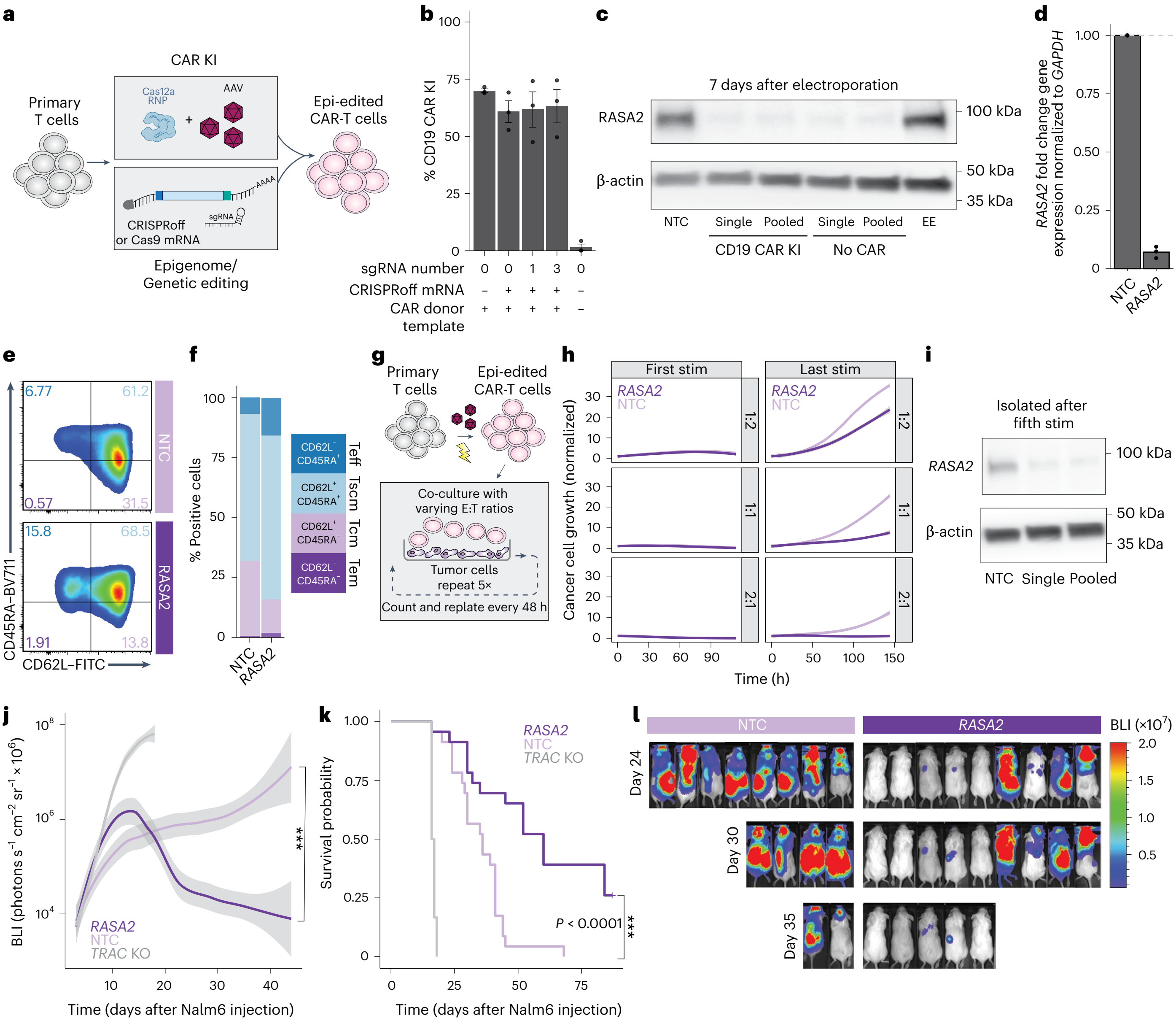

Fig. 5 | An integrated approach for simultaneous epigenetic and genetic engineering of CAR-T cells. a, Schematic of a method for simultaneously generating epigenetically and genetically engineered CAR-T cells using Cas12a RNP for CAR KI. **b**, Graph comparing KI efficiency of CD19-specific CAR with no mRNA present or CRISPRoff mRNA used in combination with an NTC or sgRNA targeting *RASA2*. CRISPRoff was electroporated with either one sgRNA or a pool of three sgRNAs targeting the *RASA2* TSS. Conditions noted as 0 sgRNA indicate an NTC (mean ± s.e.m.; $n = 3$ donors). **c**, Western blot comparison of *RASA2* silencing with CRISPRoff with or without a CD19-specific *TRAC* CAR. CRISPRoff was coelectroporated with either a single NTC sgRNA, a single sgRNA targeting *RASA2* or a pool of three sgRNAs targeting *RASA2*. Data are representative of one donor. **d**, Cells with a *TRAC* CAR KI and *RASA2* KD or NTC were collected at 7 days after electroporation for RT–qPCR. Transcript levels show *RASA2* normalized to *GAPDH*, relative to the NTC ($n = 3$ donors). **e**,**f**, T cell immunophenotypes on day 7 based on CD45RA and CD62L expression, measured by flow cytometry and shown as either raw flow plots (**e**) or bar charts (**f**). Data are representative of one donor. **g**, Schematic of the repetitive stimulation assay to examine the functional efficacy of *RASA2* epi-silenced CAR-T cells. **h**, Graphs show CAR-T cell cytotoxicity according to Incucyte analysis after five repetitive stimulations with target cancer cells. Dark-purple lines indicate *RASA2* epi-edited CAR-T cells and light-purple lines are control-edited CAR-T cells. The line is the mean and shaded areas depict the 95% confidence interval for technical replicates across three independent donors ($n = 3$ donors). Each row represents an E:T ratio (top, 1:2; middle, 1:1; bottom, 2:1). **i**, Western blot for *RASA2* expression in CAR-T cells that were treated with either one sgRNA or a pool of three sgRNAs targeting *RASA2*, which were isolated after the fifth repetitive stimulation. Data are representative of one donor. **j**, NSG mice were injected with 0.5×10^6 Nalm6 cells followed 4 days later by 0.1×10^6 *RASA2* epi-silenced CD19-specific CAR-T cells or CD19-specific CAR-T cells treated with an NTC. Tumor burden was monitored by BLI. The line is the mean and shaded areas depict the 95% confidence interval across replicates ($n = 4$–5 mice per group across four independent experiments and four donors, for a total of 23 mice per group; two-sided Mann–Whitney *U*-test: $P = 0.0004$). Individual experiments are shown in Supplementary Fig. 9. **k**, Survival of *RASA2* epi-silenced CD19-CAR-T cells shown in **i**. Survival curves were compared using a log-rank test ($P = 2.2 \times 10^{-16}$). **l**, Representative images for the mice shown in **k**.

epi-edited CAR-T cells can maintain stable target gene silencing even through multiple rounds of successful antigen-positive cancer cell killing, enabling functional enhancement of CAR-T cells through silencing of 'checkpoint' genes without the need for multiplexed gene cleavage.

Discussion

We established an all-RNA CRISPR-based epigenetic editing platform for multiplexed primary human T cell programming. Previous work with CRISPRoff and related systems have demonstrated robust and stable epigenetic silencing in cell lines such as HEK293T cells[24,26,27,62]. While important for initial optimization, these cell lines demonstrate a variety of abnormalities such as endogenously low levels of the TET enzymes that reverse DNA methylation from CpG dinucleotides[63]. We now show that, in primary human T cells, which express high levels of TET2 and TET3 enzymes[64], silencing of endogenous genes with and without a CGI is stable following only transient expression of CRISPRoff[62]. This approach is highly specific to the target loci and durable through multiple T cell activations, numerous cell divisions and transfer in vivo. It is also compatible with massive multiplexing, eliminating the cytotoxicity and genotoxicities observed with genome engineering using nuclease-active Cas9 or base-editing approaches[65]. Critically, because CRISPRon and CRISPRoff need to be delivered only transiently to exert stable effects, they circumvent the immunogenicity of constitutive Cas protein required for altering expression at the RNA level through CRISPRa, CRISPRi or RNA-targeting Cas species[21–23].

Transient delivery of CRISPRoff is critical to prevent host rejection of the ultimate cellular products. Here, we optimized mRNA delivery for CRISPRoff by combining cap structure, codon optimization and base modifications to substantially increase mRNA potency. This system enables complete silencing of five concurrent targets in this study and we expect this number could be greatly expanded. Our approach is compatible with current electroporation-based manufacturing processes and the required GMP reagents and equipment. We expect that, for any CRISPR technology, there exists the risk of off-targets and this risk should be carefully evaluated for each unique sgRNA and gene target in any clinical program.

Additionally, we developed an all-RNA platform for CRISPRon that can remove endogenous methylation from the TSDR of *FOXP3*. While prior studies with CRISPRon in HEK293T cells have reactivated genes that were previously silenced by CRISPRoff, here, we targeted a critical endogenously methylated noncoding region to establish stable de novo expression of FOXP3 in a $CD4^+$ Tconv cell population over time. These data contrast with a previous attempt to deliver plasmid encoding dCas9 fused to TET1 and an sgRNA targeting the TSDR to primary human T cells, which resulted in rapid remethylation of the TSDR over time, even when *FOXP3*-expressing clones were isolated[51]. In this study, we only demonstrated CRISPRon activity at one enhancer; however, future efforts may focus on establishing the generalizability of this tool across diverse genomic elements. We anticipate that multiplexing with both CRISPRon and CRISPRoff will provide a foundation for systematic reprogramming of chromatin architecture in primary human cells.

Lastly, we demonstrated durable silencing for a variety of clinically relevant T cell genes and developed two strategies compatible with clinical translation that combine CRISPRoff silencing of *RASA2* with targeted *TRAC* locus CAR KI, using either truncated sgRNAs or orthogonal Cas species to circumvent the issue of guide swapping. There are many alternative therapeutic targets and a remaining question is how generalizable CRISPRoff-mediated gene silencing will be across different genomic loci with varying amounts of CpG dinucleotides. We show robust and durable silencing at promoter regions both with and without well-defined CGIs. However, silencing at two of the five non-CGI genes exhibited reduced stability. In addition, regulation of gene expression can be complex and driven by multiple regulatory elements in a cell-state-specific manner. Additional studies are needed to establish rules for stable silencing or activation across diverse genomic loci and cell states, as well as the requirements for CpG content and genomic context. We anticipate that large-scale functional genomics screens across promoters, enhancers and other regulatory regions will be enabled by this platform and could shed light on the rules governing stable versus metastable gene silencing. These studies can also provide important information for mapping and dissecting the functions of noncoding elements in the genome, which can lead to novel therapeutic strategies as with the context-specific enhancer targeted in therapies for sickle cell disease and β-thalassemia recently approved by the US Food and Drug Administration[66]. We expect that leveraging CRISPRoff and CRISPRon will offer insights into gene regulation, epigenetic landscapes and the intricacies of cellular differentiation. Moreover, leveraging these technologies in primary human cells paves the way for the next wave of advanced cellular products with finely tuned control of the epigenetic state to improve the potency, durability and safety of engineered cellular therapies.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41587-025-02856-w.

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